ASHAMitra — The 11-Step Decision Pipeline

> The on-device clinical "brain". Source: `lib/core/services/rule_executor.dart` + `lib/core/services/layers/`. Companion docs: clinical modules · sign-off rules · overview + backend.

The pipeline takes a patient's **answers + measured vitals** and runs them through **11 layers in a fixed order**, producing one `DecisionOutput`: the band (🟢/🟡/🔴), referral, action card, follow-up, and a full audit trace. It runs entirely **on the phone, offline**.

🟢 In easy words

Picture a **careful nurse taking the patient through 11 safety gates**, one after another. The answers + measurements go in one end; a clear decision (a colour + what to do) comes out the other.

1. **Form check** — is everything filled in properly?
2. **Sense check** — any impossible entry (like oxygen 105%)?
3. **Right checklist** — newborn form for a newborn, etc.
4. **Missing measurement?** — note what wasn't measured, and stay extra careful.
5. **Rulebook seal** — is the rulebook the real, untouched one?
6. **Main decision** — run the danger-sign rules → first colour.
7. **Danger score** — add up points for each problem.
8. **Background** — very new baby / risky-age mother / past red case → be stricter.
9. **Safety net** — one last double-check; bump up if anything still looks unsafe.
10. **Where + how fast** — which facility, how urgent, call 108?
11. **Final card** — show the worker the colour, the action, and the reason.

The golden rule

> **The pipeline can only escalate, never downgrade.** Once any RED-tier rule fires, RED is *locked* — no later layer can lower it. For a frontline worker, over-warning is always safer than missing a real danger.

Layers **1–5** are gatekeepers (a blocking error here halts the pipeline with band `UNKNOWN` / `pipelineBlocked`). Layers **6–11** produce and refine the decision.

Layer 1 — Input Validator

File: `layers/input_validator.dart` · **Job:** make sure the form is usable before anything runs.

Blocks on: empty `moduleId`, unknown module, empty answers, a null/ wrong-typed answer (must be bool or string), a non-numeric or negative vital. Warns (non-blocking) on: empty `caseId`, a null vital (that vital's numeric rules are simply skipped).

Layer 2 — Contradiction Checker

File: `layers/contradiction_checker.dart` · **Job:** catch impossible or self-contradicting input.

Hard-blocks (physically impossible vitals): SpO_2 > 100%, temp > 45 °C, respiratory rate > 120/min, systolic BP > 300, **diastolic ≥ systolic**. **Warns only (clinically odd but possible):** e.g. lethargy without poor feeding, cyanosis without breathing difficulty, severe dehydration without diarrhoea, bleeding with normal fetal movement.

Layer 3 — Age / Module Validator

File: `layers/age_module_validator.dart` · **Job:** right checklist for the right patient.

Age bounds: newborn **0–28 d**, child **61 d–5 y**, immunisation **0–16 y**; pregnancy & postpartum have no age bound but **require female sex** (male → block). Mismatch suggests the correct module. Warns if no age is given for an age-sensitive module.

Layer 4 — Required Vital Checker

File: `layers/required_vital_checker.dart` · **Job:** note which critical vitals are missing.

Per module it knows the **RED-tier vitals** (e.g. newborn: SpO_2, RR, temp; pregnancy: systolic, diastolic, Hb; postpartum: temp, Hb). Missing ones **don't block** — they set `hasBlockingGaps`, which Layer 8 uses to escalate conservatively (you can't call a patient "safe" if you never measured the dangerous vital).

Layer 5 — Protocol Hash Verify

File: `layers/protocol_hash_verifier.dart` · **Job:** tamper seal on the rulebook.

Computes a SHA-256 of `asha_engine.json`. First load registers the hash; later loads compare against it — a mismatch within a session halts the pipeline (`HASH_001`). The hash is also written into the audit trace.

Layer 6 — Rule Engine (the heart)

File: `layers/rule_engine.dart` · **Job:** run the module's rules and set the provisional band.

Four stages, in order: 1. **hard_stop** — any one matching answer → `redLock = true`, band RED. 2. **combination** — all conditions of a combo true → RED (locks) or YELLOW. 3. **numeric** — vital thresholds (only if vitals were given) → RED or YELLOW. 4. **yellow** — single smaller signs → YELLOW. **Skipped entirely once RED is locked.**

Graded answers: an answer of "*mild/intermittent*" or "*unsure*" on a danger sign forces at least YELLOW (never let an uncertain danger sign pass as GREEN). Also computes the raw risk score for the audit trail. **Outputs:** `redLock`, `provisionalBand`, `triggeredRules`, suspected conditions, danger signs, `winningRule` (drives the action card), and a per-rule trace.

Layer 7 — Severity Scoring

File: `layers/severity_scoring_engine.dart` · **Job:** a weighted danger score (advisory — never downgrades RED).

`total = answer-weights + vital-penalties`. Only the **highest penalty per vital** counts:

Vital	Penalty ladder
SpO_2	< 90 → +4 · < 95 → +2
Respiratory rate	> 60 → +3 · > 50 → +2
Temperature	> 40 → +3 · > 38.5 → +2 · > 37.5 → +1
Systolic BP	≥ 160 → +4 · ≥ 140 → +2
Diastolic BP	≥ 110 → +3 · ≥ 90 → +2
Haemoglobin	< 7 → +4 · < 10 → +2
MUAC	< 11.5 → +4 · < 12.5 → +2

The total maps to a score-band via the module's thresholds and a risk level (LOW / MODERATE / HIGH / CRITICAL).

Layer 8 — Adaptive Risk

File: `layers/adaptive_risk_engine.dart` · **Job:** raise the band based on context. **Never downgrades RED** (returns immediately if `redLock`).

Trigger	Effect	Code
Newborn < 7 days	YELLOW → RED	ADAPT_D_001

Trigger	Effect	Code
Newborn weight < 2 kg	GREEN → YELLOW	ADAPT_D_002
Infant < 3 months (child module)	YELLOW → RED	ADAPT_D_003
Pregnancy age < 18 or > 35	GREEN → YELLOW	ADAPT_D_004
Any prior RED outcome	GREEN → YELLOW	ADAPT_H_001
High-Risk-Pregnancy flag	GREEN → YELLOW	ADAPT_H_002
≥ 2 missed ANC visits	GREEN → YELLOW	ADAPT_H_003
Last visit RED < 48 h ago	YELLOW → RED	ADAPT_H_004
RED-tier vital missing	GREEN → YELLOW	ADAPT_V_001
RED-tier vital missing	YELLOW → RED	ADAPT_V_002
Score-band RED	YELLOW → RED	ADAPT_S_001
Score-band YELLOW	GREEN → YELLOW	ADAPT_S_002

Layer 9 — Safety Escalation

File: `layers/safety_escalation_layer.dart` · **Job:** final safety net (5 checks).

1. **Sign-off-pending note** — if a fired rule is draft, attach a `SAFETY_A_001` advisory (no band change). See SIGNOFF_RULES.md.
2. **Emergency cross-sweep** — re-run the **emergency** module's hard-stops against the answers even in a non-emergency module; any hit → RED.
3. **RED-lock floor** — if `redLock` was set but band somehow isn't RED, force RED.
4. **Referral sanity** — if band is RED but referral is blank, default to "FRU / SNCU / DH immediately".
5. **GREEN-with-danger-signs** — if band is GREEN yet danger signs exist, bump to YELLOW.

Layer 10 — Referral Decision

File: `layers/referral_decision_engine.dart` · **Job:** where to send, how fast, follow-up.

Band	Urgency	Max delay	Transport
🔴 RED	Immediate	30 min	"Call 108 now" (or "keep stable" if already called)
🟡 YELLOW	Within 24 hours	1440 min	Arrange transport (+ distance estimate)
🟢 GREEN	Routine	0	None

Facility depends on module+band (e.g. newborn-RED → SNCU/FRU; pregnancy-GREEN → routine ANC at PHC). Travel time is estimated at 30 km/h rural road speed. Follow-up: RED-refused recheck in 4 h, YELLOW in 24 h.

Layer 11 — Explainable Output

File: `layers/explainable_output.dart` · **Job:** assemble the final answer the worker sees.

Builds the bilingual (Bengali + English) **action card** ("🔴 জরুরি …" / "🟡 সতর্কতা …" / "🟢 স্বাভাবিক …") plus referral, suspected conditions, danger signs, severity breakdown, adaptive adjustments, safety flags, missing vitals, the protocol hash, and the full rule-by-rule trace — so any case can be replayed/audited later.

Supporting services

- `vitals_extractor.dart` — pulls BP, temperature (auto-converts °F→°C when value > 45), MUAC, SpO_2, weight, respiratory rate, and haemoglobin out of free speech; normalises Bengali digits; and can speak an immediate danger alert.
- `offline_brain.dart` — chooses the next most-urgent question to ask (priority: hard-stop 100 > combination 60 > yellow 30 > base 10, with a +60 boost if answering it would complete a RED combination) and fires combination/single-sign alerts; ends the session at 3+ confirmed danger signs.

One worked example

Newborn, worker enters "*fever 37.8°C*", baby is 5 days old: 1. L1–L5 pass. 2. **L6** — `NB-002` (any newborn fever) and `NB-VITAL-003` (≥37.5 °C) both fire → **RED locked**. 3. **L8** — newborn < 7 days, but it's already RED, so unchanged. 4. **L10** — SNCU/FRU, immediate, "call 108". 5. **L11** — red card: "*🔴 জরুরি — এখনই SNCU-তে রেফার করুন।*"